

Executive Summary

Project Overview

This project analyzes the product line profitability and margin performance of Nassau Candy Distributor using sales, cost, units and gross profit data. The objective was to identify profitable products, margin-risk products, major division contributors, profit concentration and cost-related issues. An interactive Streamlit dashboard was also developed to present the analysis and allow users to explore the results using filters.

Key Findings

The analysis showed that the **Chocolate division** was the major contributor to business performance, accounting for **92.88% of total revenue and 95.06% of total gross profit**. Profitability was highly concentrated, with only **5 out of 15 products (33.33%)** contributing approximately **95.06% of total gross profit**.

Wonka Bar - Scrumdiddlyumptious generated the highest gross profit of **\$19,357.50**, while **Wonka Bar - Nutty Crunch Surprise** recorded a gross margin of **71.35%** among the top five products. In contrast, **Kazookles** had a gross margin of only **7.69%**, with cost representing **92.31% of sales**, making it an important product for margin review.

Regional margins were relatively consistent, ranging from **65.47% to 66.43%**, while monthly analysis showed stronger sales and gross profit performance during **2025**, particularly toward the end of the year.

Recommendations

The analysis suggests continued monitoring of the core high-profit products, review of high-cost and low-margin products and closer examination of pricing and sourcing costs where margins are weak. The Other division can also be reviewed for opportunities to improve cost efficiency. The Streamlit dashboard can be used to monitor these profitability indicators interactively.

Tools Used

Python, Pandas, Jupyter Notebook and Streamlit were used for data analysis and dashboard development.

Project Outcome

The project provides a structured view of product and division profitability and demonstrates how data analysis can support business decisions related to product performance, cost management and margin improvement.